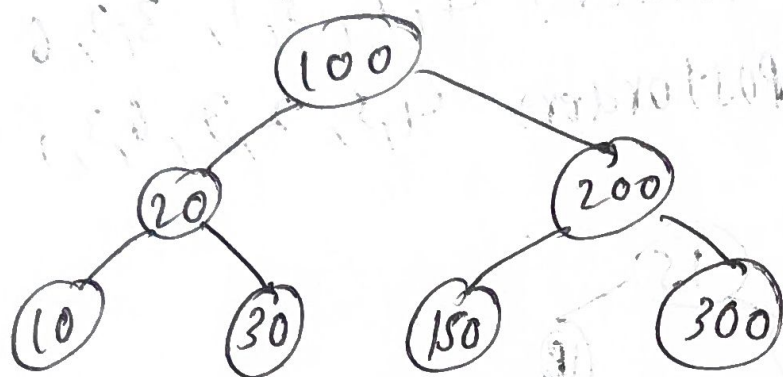


C03: Tree Traversal

Q1.

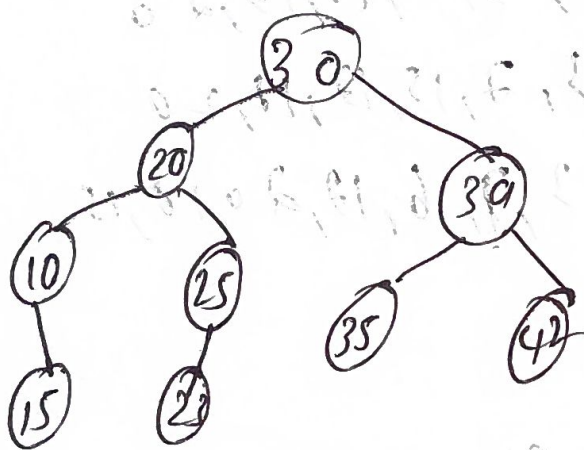


pre order: 100, 20, 10, 30, 200, 150, 300

Inorder: 10, 20, 30, 100, 150, 200, 300

postorder: 10, 30, 20, 150, 300, 200, 100

Q.2



Preorder: 30, 20, 10, 15, 25, 23, 39, 35, 42

Inorder: 15, 10, 20, 23, 25, 30, 35, 39, 42

postorder: 15, 10, 23, 25, 20, 35, 42, 39, 30

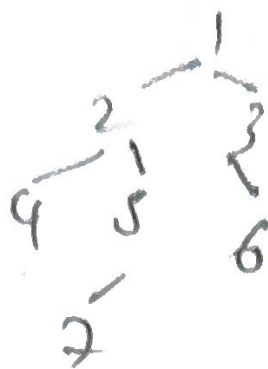
Q3: BST built by inserting 7, 5, 1, 8, 3, 6, 0, 9, 4, 2

pre order: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Inorder: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Post order: 0, 2, 4, 3, 1, 6, 5, 9, 8, 7

Q5.

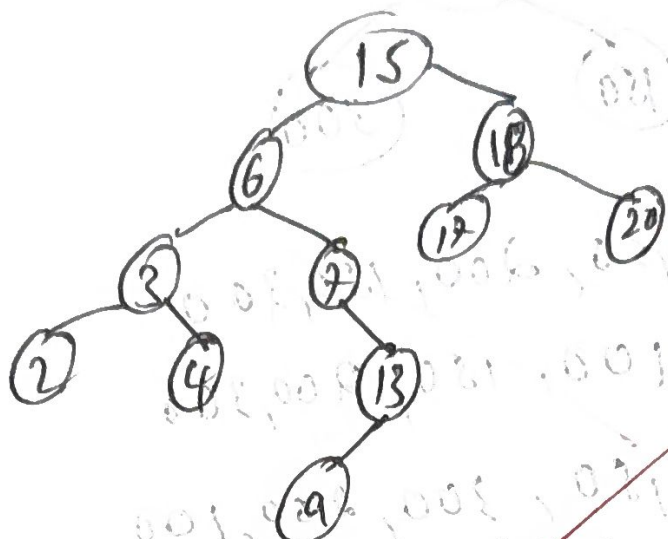


preorder: 1, 2, 4, 5, 3, 6, 2

Inorder: 4, 2, 5, 1, 3, 2, 6

postorder: 4, 5, 2, 3, 6, 3, 1

Q6.

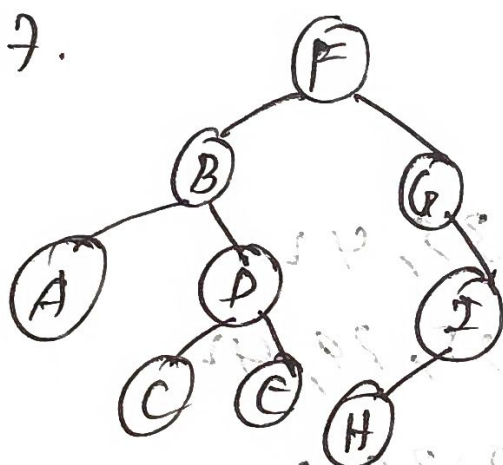


Preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 12, 20

Inorder: 2, 3, 4, 6, 9, 13, 7, 15, 12, 18, 20

Postorder: 2, 4, 3, 9, 13, 7, 6, 12, 20, 18, 15

Q7.



In order:

A, B, C, D, E, F, G, H, I

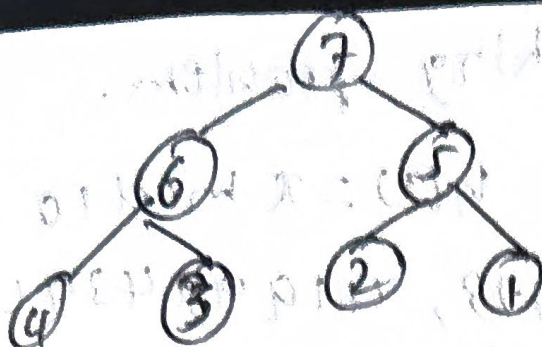
option 3

Q8. Postorder: D, F, E, B, H, G, C, A → option 3

Reconstructed the tree: A is root, B/C is

Children: B → D, E; E → F; C → G; G → H;

Q9.

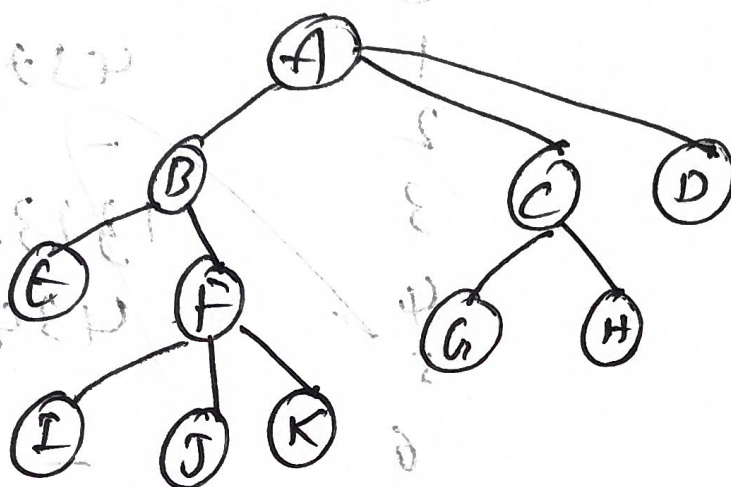


preorder: 7, 6, 4, 3, 5, 2, 1

Inorder: 4, 6, 3, 7, 2, 5, 1

postorder: 4, 3, 6, 2, 1, 5, 7

Q10:



pre order: A, B, E, F, I, J, K, C, G, H, D

postorder: E, I, J, K, F, B, G, H, C, D, A

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